

1)

- a) The A/B designates two pumps, a primary pump and a backup pump. We choose to have 2 pumps because 1) pumps are cheap and 2) having the entire process go down because of a pump failure is bad, therefore having a secondary pump to switch to in the case of a pump failure is good practice.
- b) Reactor A/B would be used when there is a lifetime for the reactor, such as a catalyst that needs to be replaced every X days or a bed that is filled with contaminant and needs to be replaced with new material. Having two reactors allows you to switch from one reactor to another while you change out parts in the old reactor, giving better uptime for the process.
- c) Building a scale model is crucial for visualizing the plant and can reveal issues with piping, physical spaces, and power/utility connections, along with operator space for OSHA regulations.
- d) It is important to know the material something is made of in case there are potential conflicts with contained chemicals or safety hazards. In addition it is helpful to know the material for material properties, such as melting point and thermal conductivity.
- e) It is to be named something different, such as V-119. This is because even though the function is the same, the specifications are different and therefore could cause issues in the future.
- f) If the vessel is identical, a small notation should be made to the name to designate its replacement, but the number should remain the same, i.e. V-101a.
- g) The PNID, as it should have the specifications on the exact parts of the level controller.

2)

- a) Stream table
- b) Block plant diagram
- c) Block process diagram
- d) PNID
- e) PNID

3)

- a) E-208 is a heat exchanger specifically for the bottom flow of the distillation column. More than likely it is to ensure wastewater is of an acceptable temperature, as high temperature fluids could be prone to flashing, reacting with other waste, or corroding infrastructure downstream. In addition, the local municipality could have local regulations designating the maximum allowable temperature of the wastewater.
- b) Mps stands for medium pressure steam
- c) The stream leaving the reactor is 365 degrees C
- d) 130 kmol/hr

- e) The surface area of E-204 is 22.0 m^2 , is made of carbon steel, and is rated for a Q of 2490 MJ/hr
- f) The recycle stream 13 is included in the process design so that unreacted methanol in the distillate can be sent back through the process.
- g) There is no overhead stream from V-202 because it is fully liquid, insinuating that E-205 is a complete condenser and condenses all the gas from the distillation column into a liquid.
- h) In E-202, the output stream from the reactor, which is hot vapor with around 40% water and dimethyl ether and 20% unreacted methanol, is being used to pre-heat the incoming stream of the reactor, which has a composition of roughly all methanol. This means that the output stream is losing heat, transferring it to the incoming stream. Since there are no heating elements labeled in the reactor, we can assume that the reaction is extremely exothermic. By using the output stream of the reactor to preheat the input stream, you lower the temperature of your output stream (which you need to do for your distillation column) and increase the temperature of your input stream, encouraging the reaction to proceed in the reactor.
- i) There are two distillation columns to separate the product and then separate the reactant. In T-201, the product is separated from the methanol/water/DMO mixture. The bottom product is a mixture of unreacted methanol and water. In order to separate that stream and return unreacted methanol to the process, it is separated with another distillation column.